

UAV OPERATION

BRICS FUTURE Hi-Tech Skill
Championship – Ekaterinburg, Russia

Presented by
SureshKumar Balasubramanian



MODULE A

- Farmers face difficulties in transporting produce and materials across large fields.
- Manual transport is time-consuming and labor-intensive.
- Objective: Design and development of a cargo transportation unit for agriculture



Component Details

- Flight Controller: Pixracer
- Receiver: ELRS
- Transmitter: Radiomaster TX12 ELRS
- Motors: 2300KV BLDC (x4)
- Propellers: 5-inch Triblade
- Servo: 5V Metal Gear Servo
- Gripper: 3D Printed



Electrical Schematic

Electrical schematic diagram(showing connections for FC, motors, ESCs, servo, and power).

ELRS
RECIVER IN



SERVO OUT
AUX 5

MAIN OUT FOR
MOTORS
Aux 1 - Aux 4

POWER INPUT

USB FOR
CONNECTION AND
CALIBRATION

Working Principle

- Drone hovers over cargo location.
- Servo gripper closes to pick up payload via AUX channel 5 control.
- Drone navigates to drop point.
- Servo opens to release payload.
- Demonstrates remote-controlled precision and stability.



Flight Controller Setup – Pixracer

- Configured through Mission Planner.
- AUX 5 output assigned to servo channel.
- Servo end-points calibrated.
- ELRS Receiver paired with Radiomaster TX12 for reliable signal control.



Servo Gripper Mechanism

- 3D printed mechanical hook attached to drone base.
- 5V Metal Gear Servo provides actuation.
- Controlled via transmitter switch mapped to Channel 5.
- Lightweight and efficient design to handle moderate payloads.



MODULE B

- The powerlines are huge source of electricity us so for the maintance of the powerlines we are using drones for it
- Objective : Maintance of powerlines using UAV



Project Objective

- Design and build a UAV for power line inspection and light maintenance tasks.
- Integrate a servo-controlled manipulator for simple mechanical repairs or cleaning.
- Optimize flight stability using Pixracer FC and ELRS control system.
- Enable safe remote operation near high-voltage environments.



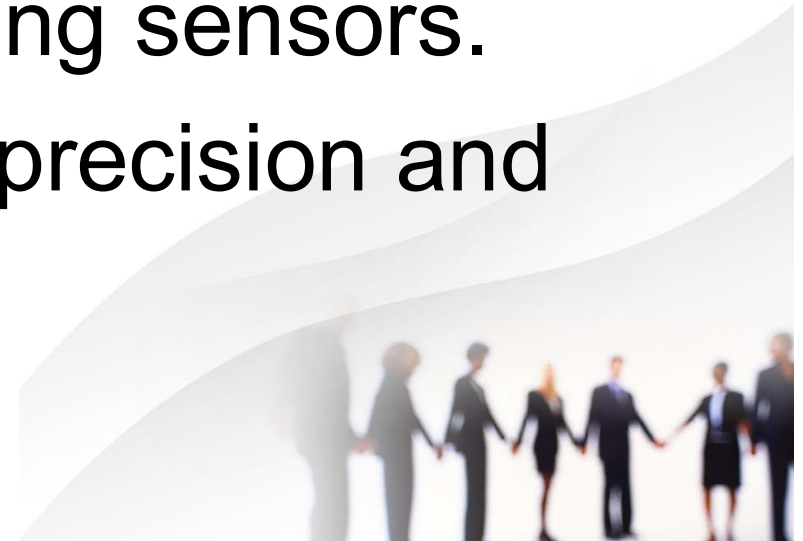
Component Details

- Flight Controller: Pixracer
- Receiver: ELRS
- Transmitter: Radiomaster TX12 ELRS
- Motors: 2300KV BLDC (x4)
- Propellers: 5-inch Triblade
- Servo: 5V Metal Gear Servo
- Drone Arm: 3D Printed



Servo Arm Mechanism

- 3D-printed lightweight servo arm mounted under UAV.
- Controlled via transmitter switch mapped to Channel 5.
- Designed for minor maintenance like brushing insulators or placing sensors.
- Metal gear servo provides precision and strength.



Demonstration Summary

- Demonstration includes lifting and dropping a payload within the farm field.
- Showcases control, stability, and payload handling.
- Smooth servo operation and safe Landing on the powerlines with removal of obstacles are demonstrated.



Safety & Maintenance

- • Pre-flight check: Verify all connections and propellers.
- • Maintain 10-meter safety zone during operation.
- • Regular calibration of flight controller.
- • Inspect gripper mechanism before each flight.



Documentation Overview

- User manual for operation and maintenance.
- 3D model of the servo Arm
- Wiring schematic and assembly steps documented.
- Test results recorded.
- Data to be used for further system optimization.



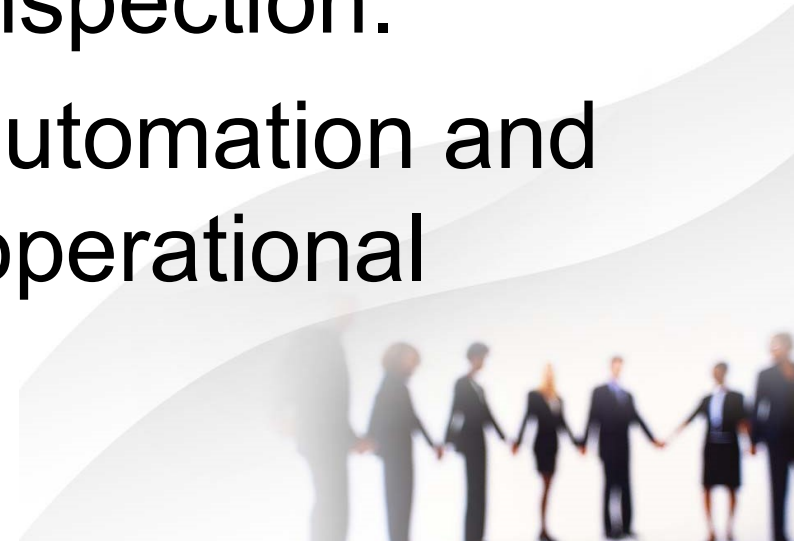
Future Development

- Integrate GPS for Position Hold.
- Enable autonomous waypoint missions.
- Develop lightweight frame for improved efficiency.
- Add telemetry for real-time monitoring.
- Increase payload capacity.



Conclusion

- The cargo transport drone provides a practical, efficient, and scalable solution for farm logistics.
- The project demonstrates the integration of mechatronic systems with UAV technology for Powerline Inspection.
- Further advancements in automation and GPS control will enhance operational reliability.



REFERENCES

- <https://px4.io/>
- <https://qgroundcontrol.com/>
- <https://github.com/mavlink/qgroundcontrol>
- <https://github.com/ArduPilot/ardupilot>



Acknowledgment

- Presented by:
SureshKumarBalasubramanian
- Representing: India
- Event: Hi-Tech Skill Championship,
Ekaterinburg, Russia
- Special thanks to mentors, organizers,
and supporting institutions.

